we need to use the line-point distance, and others when the distance to an extreme is what matters.

In turn, when edges are drawn with straight-line segments, this forces us to check three distances: if the distances between the two segment's endpoints (S, E in figure 15.28) and the vertex's center (let's call it C) is smaller than the vertex radius, then we certainly have an intersection; even if these distances are larger, though, the vertex could intersect the edge somewhere in the middle, so we need to check the distance between the line passing through the segment and the vertex's center, and that the point of minimum distance between the vertex and the line falls inside the segment.

To find the point-to-segment distance, we can use the following formula (which we won't derive here):

$$dist(\bar{SE}, C) = \frac{\left|(E_y - S_y)\,C_x - (E_x - S_x)\,C_y + E_x S_y - E_y S_x\right|}{\sqrt{(E_x - S_x)^2 + (E_y - S_y)^2}}$$

And finally, to check if the intersection between the line passing through the segment and the line of minimum distance to a vertex (a line perpendicular to the segment passing through a vertex) falls within or outside a segment, we can use the same algorithm we developed for segments intersection, using points S and E on one side, and v and P_v (or u and P_u) on the other side: we just need to check that the multiplicator factor is between 0 and 1.

15.5 Summary